Department of Electrical and Computer Engineering

North South University

Course: CSE331L – Microprocessor Interfacing and Embedded Systems Lab

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Group Number: 3

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Greenhouse System

1. Abstract

This project presents the design and implementation of an Enhanced Greenhouse Monitoring System using the STM32F103 microcontroller. The system integrates temperature, humidity, soil moisture, and water level sensing, combined with a servo-based actuator for automatic responses. An HD44780 LCD with I²C backpack is used for real-time display of system data and status. The system supports alarm logic with hysteresis, error tracking, calibration mode, and adaptive control strategies. Both hardware interfacing and firmware optimizations ensure robustness, accuracy, and reliability for smart agriculture applications.

2. Introduction

Greenhouse automation is essential for improving plant growth by maintaining optimal environmental conditions. Traditional manual monitoring is inefficient and error-prone. This project addresses those challenges by building a microcontroller-based monitoring and control system with the following objectives:

- Monitor: Collect temperature, humidity (via DHT11), soil moisture, and water tank level (via HC-SR04).
- Control: Drive an SG90 servo motor to regulate irrigation when thresholds are breached.
- Display: Provide real-time sensor feedback and system states on a 16×2 LCD.
- Alert: Trigger alarms using LEDs and servo motion under critical conditions.

This implementation builds on STM32Cube HAL drivers, precise timers, and modular embedded C code to create a reliable greenhouse monitoring prototype.

3. Methodology

Hardware Implementation

According to the hardware documentation:
- DHT11 Sensor: Connected to PB9, configured as GPIO output switched to input for one-wire communication.
- HC-SR04 Ultrasonic: TRIG at PA9 (output), ECHO at PA8 (input).
- Soil Moisture Sensor: Analog output at PB1 (ADC1_IN9) with power gated via PA7.
- Status LEDs: PA5 = Red (dry), PA3 = Green (medium), PA0 = Blue (wet).
- Servo Motor (SG90): Driven via TIM2_CH2 on PA1, configured for 50 Hz PWM.
- LCD (I²C backpack): Communicates over PB6 (SCL) and PB7 (SDA) at 100 kHz.

Software Implementation

The firmware is developed in STM32CubeIDE with HAL libraries. Key features:

1. Sensor Drivers:
   - DHT11 bit-banging with TIM1 microsecond delays.
   - HC-SR04 with pulse width capture and overflow handling.
   - Soil sensor ADC sampling with averaging for noise reduction.

2. Control Logic:
   - Alarm logic uses hysteresis and error thresholds to avoid false triggers.
   - Servo PWM control for irrigation with safety timeout.
   - LEDs indicate soil moisture status and alarm state.

3. User Interface:
   - LCD rotates between four pages: Temperature/Humidity, Tank Distance, Soil Moisture, and Statistics.
   - System readiness checks and error messages displayed at startup.

4. System Health & Data Logging:
   - Tracks sensor errors, recovery, and min/max values.
   - Adaptive update rates: fast in alarm mode, slower during normal operation.

4. Results

- Functional Verification:
  - Sensors successfully provide real-time measurements within expected ranges.
  - Servo responds promptly under alarm conditions, ensuring irrigation control.
  - LEDs provide clear soil moisture indication (Dry/Medium/Wet).
  - LCD correctly rotates information every 8 seconds.

- Error Handling:
  - System detects and recovers from temporary sensor failures.
  - Invalid readings (e.g., ultrasonic out of range) are filtered.

- Performance:
  - DHT11 sampling interval maintained at 2 seconds.
  - Ultrasonic sensor stabilized with median filtering.
  - Soil sensor readings averaged over 5 samples for reliability.

5. Discussion

The Enhanced Greenhouse Monitoring System demonstrates robust integration of sensors, actuators, and user interface components. The error tracking and alarm logic improve reliability compared to simpler designs. Using STM32 timers ensures accurate microsecond-level

measurements critical for DHT11 and HC-SR04 operation.

Strengths:
- Multi-sensor fusion enables holistic monitoring.
- Hysteresis prevents false alarms.
- Servo integration provides active control.
- Adaptive system states (Normal, Alarm, Error, Maintenance) enhance usability.

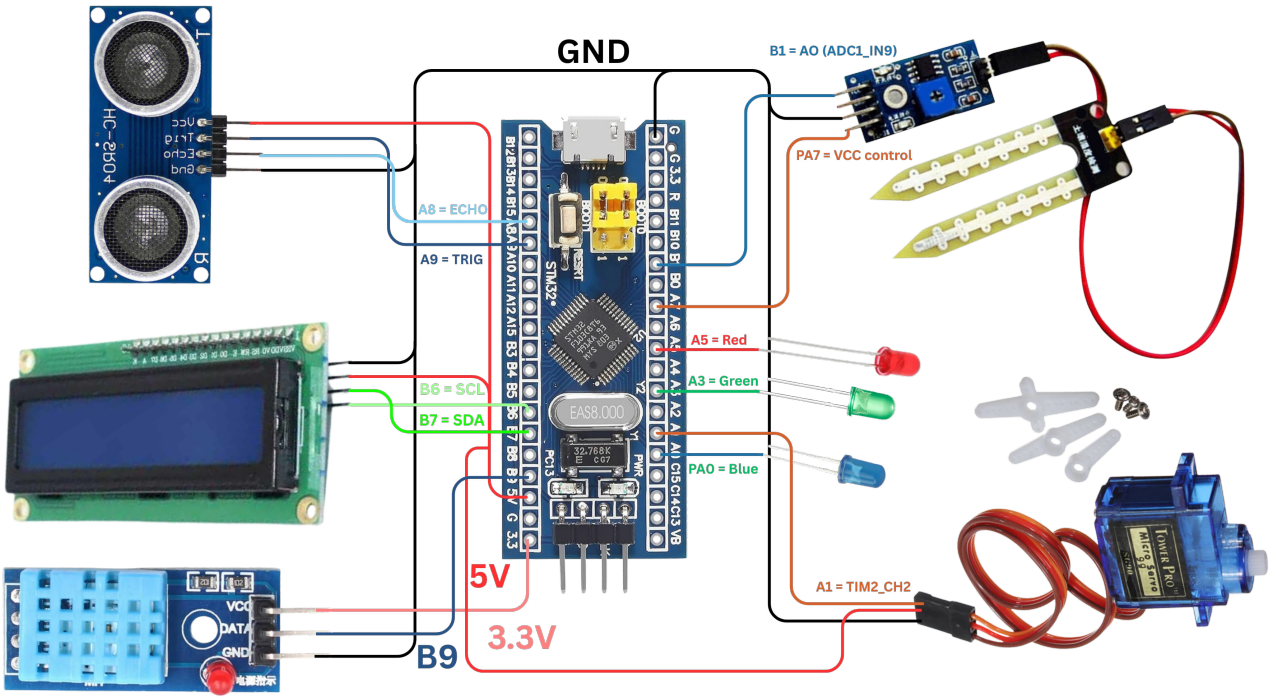
Limitations:
- DHT11 has limited accuracy (±2 °C, ±5% RH).
- Soil moisture probe corrosion remains a concern, though mitigated by power gating.
- System complexity increases firmware size, challenging low-memory controllers.

Future Work:
- Upgrade to higher-accuracy sensors (e.g., DHT22, capacitive soil sensor).
- Replace LCD with OLED for better UI.
- Add remote data logging via Wi-Fi or LoRa for IoT integration.

6. References

1. https://www.micropeta.com/video57
2. https://www.micropeta.com/video47
3. https://www.micropeta.com/video43
4. https://www.micropeta.com/video40#google_vignette
5. https://www.micropeta.com/video27

Pin Diagram


GND
B1 = AO (ADC1_IN9)
PA7 = VCC control
A8 = ECHO
A9 = TRIG
B6 = SCL
B7 = SDA
A5 = Red
A3 = Green
PA0 = Blue
A1 = TIM2_CH2
5V
3.3V
B9
HC-SR04
STM32
EA8.000
32.768K